Two-Way-ANOVA

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TWO-WAY ANOVA

Data Loading

```
data <- read.csv("G:/Lesson/Done/Research/R Language/R-Language/Self-Learning/Two_Way_ANOVA/a
data$Fertilizer <- as.factor(data$Fertilizer)
data$Irrigation <- as.factor(data$Irrigation)
```

Model Building

```
model <- aov(Yield ~ Fertilizer * Irrigation, data = data)
```

Assumption Checking

a. Normality of residuals

```r
shapiro.test(residuals(model))
```

```

	Shapiro-Wilk normality test

data:  residuals(model)
W = 0.86093, p-value = 0.003511
```

$p > 0.05 \rightarrow$ residuals are normally distributed

b. Homogeneity of variance

```r
library(car)
```

```
Warning: package 'car' was built under R version 4.4.3
```

```
Loading required package: carData
```

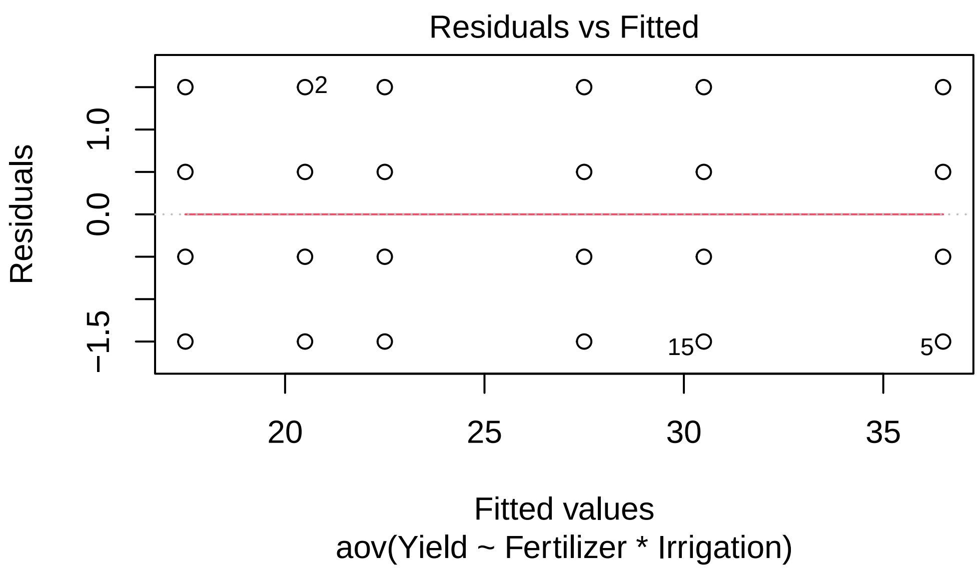
```r
leveneTest(Yield ~ Fertilizer * Irrigation, data = data)
```

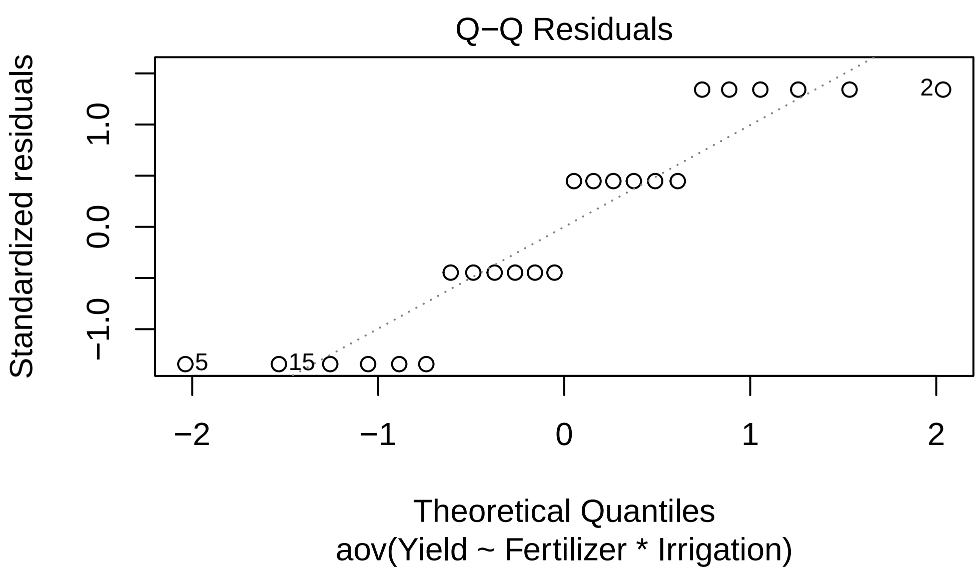
```
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  5       0      1
      18
```

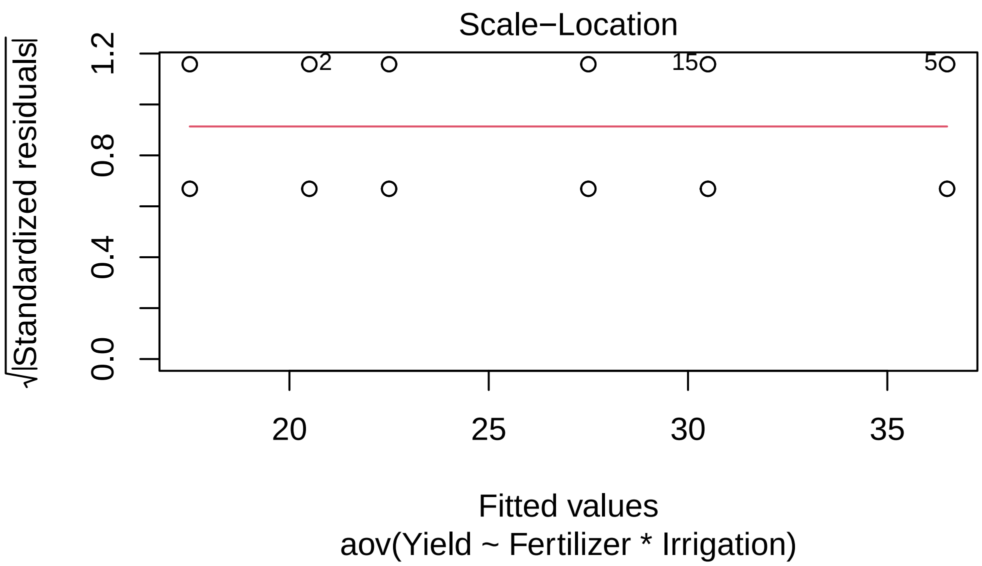
$p > 0.05 \rightarrow$ equal variance assumption satisfied

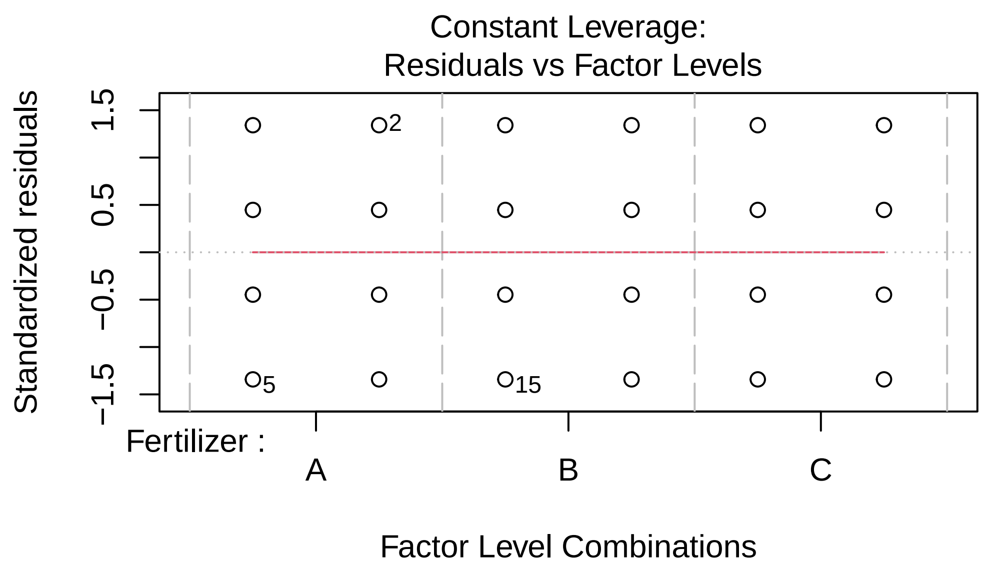
c. Residual diagnostics

```
plot(model)
```









No strong pattern → model is valid

Perform ANOVA

```r
summary(model)
```

```
                     Df Sum Sq Mean Sq F value   Pr(>F)    
Fertilizer            2  409.3   204.7   122.8 3.23e-11 ***
Irrigation            1  384.0   384.0   230.4 1.06e-11 ***
Fertilizer:Irrigation  2  196.0    98.0    58.8 1.28e-08 ***
Residuals            18   30.0     1.7                     
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Testing 3 hypotheses:

- Fertilizer effect (main effect)
- Irrigation effect (main effect)
- Fertilizer $\times$ Irrigation interaction (joint effect)

Interpretation of ANOVA Table

Main Effects-

Fertilizer effect : $p < 0.05 \rightarrow$ Fertilizer types are NOT equal. Means at least one fertilizer changes yield

Irrigation effect : $p < 0.05 \rightarrow$ Irrigation levels significantly affect yield

CRITICAL WARNING (Most beginners miss this)

If interaction is significant, main effects alone are NOT reliable because:

Effect of fertilizer changes depending on irrigation So "average fertilizer effect" becomes misleading

Interaction Effect (Most Important Part)

Fertilizer $\times$ Irrigation: $p < 0.05 \rightarrow$ Interaction exists

Correct meaning:

The effect of Fertilizer on Yield depends on Irrigation level (and vice versa)

NOT:

- “They are related”
- “Both are important”

Proper Statistical Decision Rule

Case 1: Interaction NOT significant

Interpret fertilizer main effect and Irrigation main effect separately

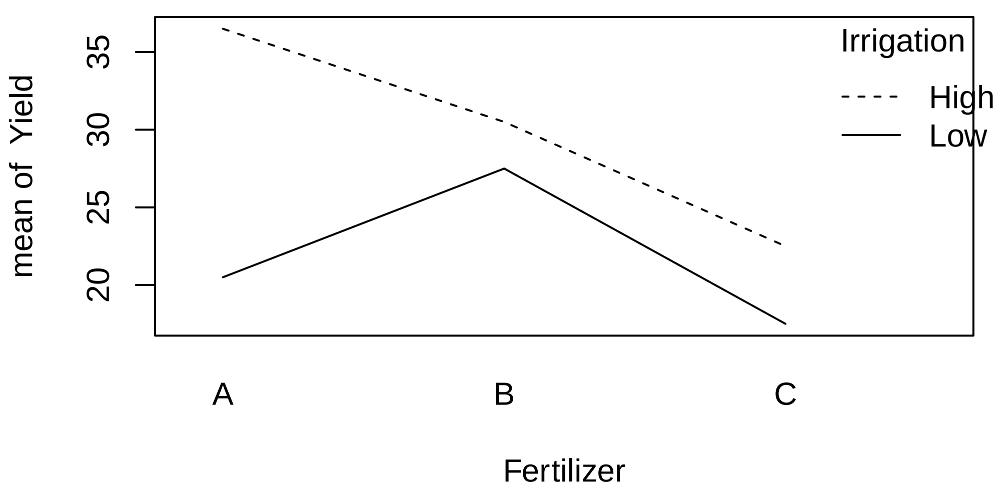
Case 2: Interaction significant

IGNORE main effects

Focus on: Interaction plots, Post-hoc within each irrigation level

Interaction Plot

```r
with(data, interaction.plot(Fertilizer, Irrigation, Yield))
```



Interpret Interaction Plot:

Step 1: Check line crossing

- Lines crossing → strong interaction
- Lines parallel → no interaction

Step 2: Interpret patterns scientifically

Example interpretation:

Fertilizer A:

High irrigation → highest yield

Low irrigation → sharp drop

That's indicate **HIGH water dependency**

Fertilizer B:

Stable across irrigation levels → drought-tolerant performance

Fertilizer C:

Low performance everywhere → generally inefficient

8. Stronger Statistical Add-ons (YOU SHOULD ADD THIS) (A) Mean Table (Very Important in research) aggregate(Yield ~ Fertilizer + Irrigation, data = data, mean) (B) Post-hoc Test (ONLY if needed)

If interaction is significant:

TukeyHSD(model)

But better approach:

Analyze simple effects instead of global Tukey blindly